Timing

EXRES
OCHAROMCS
VCHARAMCS
OCHARAMCS
HFLIP
VFLIP
DMACS
DMA
CLR1
CPURES
OCROMDTACK
SVCDTACK
OCRAMDTACK
CLK2
CLK21
CLK22
CPUCLKD
SYNCRLDV
VCNTEND
OSCANCLR
HSYNC
VBLANKD
CBLK
VSYNC
CSYNC
OBJBUFINITD
DMAINITD
DMACLR
OBJBUFCLRD
FBREF
DB0
P1HD
P1HB
X1SD
WR
N2HD
P2HD
P4HD
8HD
1VD
{HVCOUNTERS}
P256VD
P256VD
TIMD
TIM1
TIM1D
TIM2
TIM2D
TIM3D
RASB
CASB
RS1
RS2
CS1
CS2
OE1
OE2

File: timing.kicad_sch

Sprite Table

CLR1
CLK21
OBJBUFCLR
OBJBUFINIT
OSCANCLR
P256VD
DMAINIT
P1HB
P1H
P2H
FBREF
8H
HFLIP
P2HP4H
N2HP4H
DMAON
TIM
TIM3
DMACLR
CHAPAGE
OBJCS
SOBJCS
OCHARAMCS
SREAD
MREAD2
OBJSCRH
OBJSCRV
SLWR
SUWR
MLWR
MUWR
SAB[1..14]
SA[15..20]
AB[1..13]
CRW[0..3]
SPA[0..18]
SPZ17
OCHFD
FBWM1D
FBWM2D
VD[7..0]
VPD8D
HD[8..2]
HJ[0..1]
C[0..3]
CX[0..3]
DMAD
SDB[0..15]
DB[0..15]

File: sprite_table.kicad_sch

Sprite ROM and RAM

P2H
N2H
P4H
CRW[0..3]
SREAD
OCHARAMCS
OCHAROMCS
VCHARAMCS
SPA[0..18]
SPZ17
SAB1
OCHF
P2HP4H
N2HP4H
VCWENH
VCWENL
VCRDL
VCRDH
SDB[0..15]
W[0..15]

File: sprite_rom_and_ram.kicad_sch

Sprite Frame Buffers

CLR1
CLK21
CLK22
HSYNC
X1S
P2H
N2H
P256VD
P256VD
VCNTEN
WR
HFLIP
VFLIP
W[0..15]
OCHF
FBWM1
FBWM2
VD[0..7]
VPD8
HD[2..8]
HJ[0..1]
C[0..3]
CX[0..3]
CS1
OE1
RS1
OE2
RS2
CS2
OCD[0..7]

File: sprite_frame_buffers.kicad_sch

OCD[0..7]

Timing diagrams

File: timing_diagrams.kicad_sch

Sprite timing diagrams

File: sprite_timing_diagrams.kicad_sch

Spare parts on video PWB(B) 302191A

B_G2C 74LS04
B_G2B 74LS04
B_G2E 74LS04
B_G2D 74LS04
B_H11B 74LS74
B_A11A 74LS32

Spare parts are set as NC.

Spare parts on main PWB(A) 302190A

A_L2A 74LS00
A_L2D 74LS00
A_H8A 74LS07
A_H8D 74LS07
A_K1C 74LS32
A_L1C 74LS07
A_L1B 74LS07
A_L8F 74HC367
A_N3B 74LS139
GND

KiCad does not support reference designators starting with a number. Therefore the references used in the original schematics have been mirrored. 6R -> R6.
The original schematics had separate pages for the main and video PCB. They are combined here. This makes some references duplicate. To distinguish between boards A_ is added before part positions on PWB(A) and for PWB(B) B_ and so on. If there are still conflicts a "P" (Position) is added after board letter AP_, BP. Sometimes there are duplicate reference designators then "CAP" maybe added after board letter ACAP_ (CAP is for capacitor).

Tile Layers

EXRES
CLR1
CLK22
TIM1
TIM2
P1HB
X1S
P2H
P4H
P2HP4H
N2HP4H
1V
HFLIP
HSYNC
SYNCRLDV
VCNTEN
RASB
CASB
VCRDL
VCRDH
VCWENL
VCWENH
SYSFLAG
VRAMCS
SVRAMCS
VRAMCVF
VCHARAMCS
V1SCRH
V2SCRH
MLWR
MUWR
MULWR
SLWR
SUWR
MREAD2
SREAD
SA[15..17]
SAB[1..14]
AB[1..13]
1C[0..6]
2C[0..6]
SDB[0..15]
DB[0..15]

File: tile_layers.kicad_sch

Color mixer

CLK2
CBLK
PI[0..2]
FX[0..7]
1C[0..6]
2C[0..6]
OCD[0..7]
COLORRAM
AB[1..12]
MLWR
MREAD1
REDD
GREEND
BLUED
DB[0..7]

File: color_mixer.kicad_sch

IO

VSYNC
SPEAKER+
SPEAKER-
RW
IOCS
CSYNC
RED
GREEN
BLUE
MA16
AB[1..4]
DB[0..7]
EXRES
SINT
MINT16EN
DMAON
SOUND-DATA
SOUND-ON

File: io.kicad_sch

Sound

EXRES
SOUND-DATA
SOUND-ON
DB[0..7]

File: sound.kicad_sch

Sound mixer

REF2V5
PCM1
PCM2
FML
FMR
SPEECH
SPEAKER+
SPEAKER-

File: sound_mixer.kicad_sch

Only used on GX903 – Cue Brick

Battery Powered RAM

CPURES
IOCS
MREAD2
MLWR
MUWR
AB[10..1]
DB[15..0]
MA16
CPUCLK
VBLANK

File: bpram.kicad_sch

Main CPU

EXRES
CPURES
CPUCLK
TIM1
TIM3
VBLANK
DMAINIT
MINT
MINT16EN
SCPUCLK
HFLIPD
VFLIPD
FIXVRAMCS
VRAMCS
OBJCS
DMACS
COLORRAM
IOCS
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
PI[0..2]
VRAMCVF
SYSFLAG
MCOMCS
RW
MREAD1
MREAD2
MLWR
MUWR
MULWR
MA[1..23]
AB[1..13]
DB[0..15]
MA16

File: mcpu.kicad_sch

Shared Resources

CPUCLK
MCOMCS
MREAD1
MLWR
MUWR
SCOMCS
SREAD
SLWR
SUWR
AB[1..13]
SAB[1..13]
DB[0..15]
SDB[0..15]

File: shared_resources.kicad_sch

Slave CPU

EXRES
CPURES
DMAINIT
SCPUCLK
TIM2
VBLANK
SINT
TIM3
OCRAMDTACK
OCROMDTACK
SVCDTACK
CHAPAGED
MINTD
SOBJCSD
SVRAMCSD
SCOMCSD
OCHARAMCSD
OCHAROMCSD
VCHARAMCSD
SREADD
SLWRD
SUWRD
SAB[1..14]D
SDB[0..15]D
SA[15..23]D
CHAPAGE
MINT
SOBJCS
SVRAMCS
SCOMCS
OCHARAMCS
OCHAROMCS
VCHARAMCS
SREAD
SLWR
SUWR
SAB[1..14]
SDB[0..15]
SA[15..23]

File: scpu.kicad_sch

Fixed Layer

CLK21
{HVCOUNTERS}
P1HB
P2H
P4H
HFLIP
FIXVRAMCS
MLWR
MUWR
MREAD2
AB[1..13]
DB[0..15]
FX[0..7]

File: fixed_layer.kicad_sch

There are several games and variations on the twin 16 platform:
- Main Board PWB(A)
- Video Board PWB(B)
- Additional ROM Board, if not ROM's mounted directly on video board.
- Battery powered RAM board

	Id	PWB(A)	PWB(B)	PWB(C)
Dark Adventure	GX687	302190A	302191A	–
Gradius 2	GX785	302190A	302191A	–
Vulcan Venture	GX785	302190A	302191A	–
Gradius 2	GX785	302190B	302191B	402145A
Hard Puncher	GX870	302190B	302191B	402145A
M.I.A	GX808	302190B	302191B	402145A
Cue Brick	GX903	302190B	302191B	–

Cue Brick has an additional Battery Powered RAM Board:
PWB451803

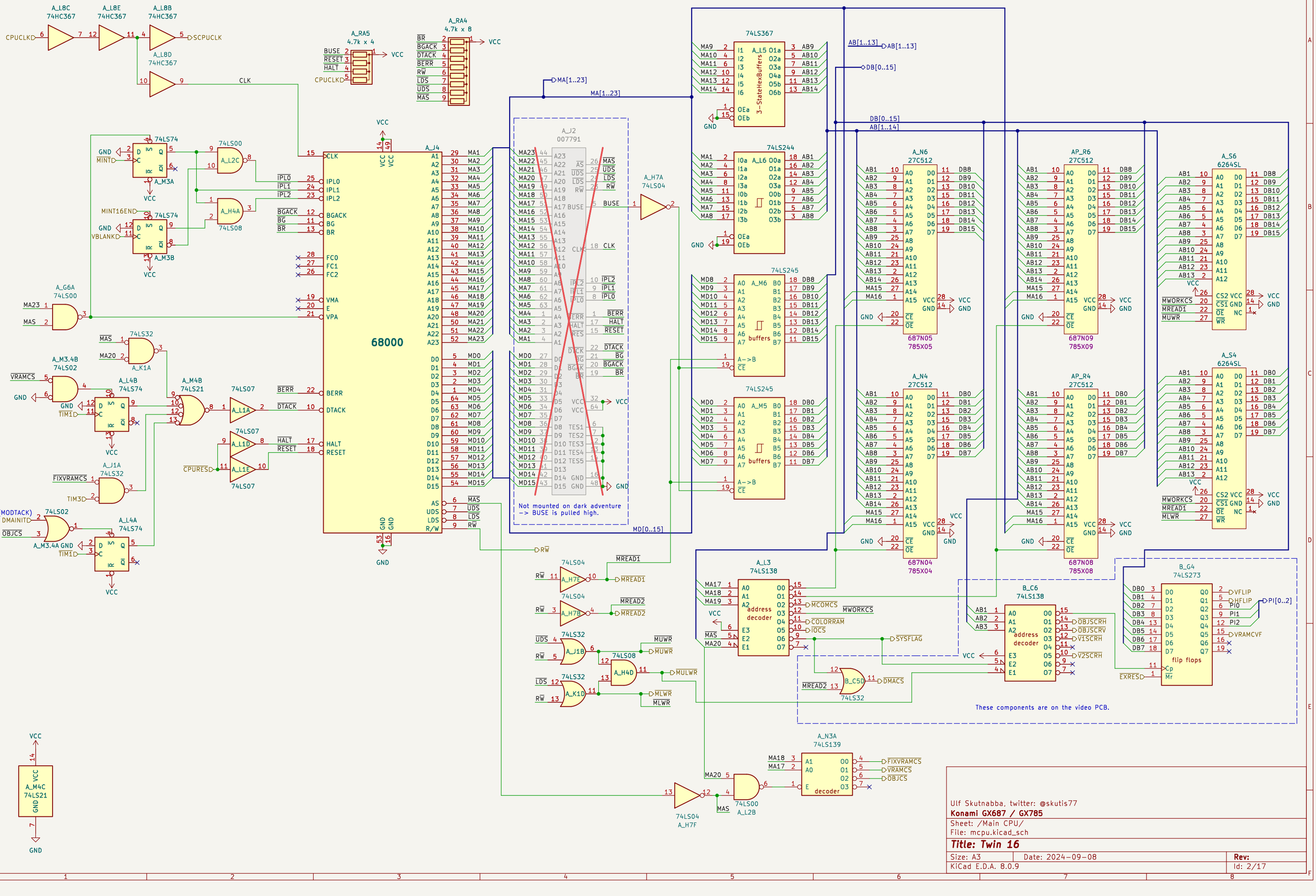
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /
File: darkadv.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 1/17


A_L8C 74HC367
A_L8E 74HC367
A_L8B 74HC367
A_L8D 74HC367
CPUCLKD
SCPUCLK
CLK
A_RA5 4.7k x 4
BUSE
RESET
HALT
CPUCLKD
VCC
A_RA4 4.7k x 8
BR
BGACK
DTACK
BERR
RW
LDS
UDS
MAS
MA[1..23]
A_J2 007791
Not mounted on dark adventure -> BUSE is pulled high.
MD[0..15]
74LS367
3-StateHexBuffers
AB[1..13]
DB[0..15]
74LS244
74LS245
buffers
A_N6 27C512
687N05 785X05
AP_R6 27C512
687N09 785X09
A_N4 27C512
687N04 785X04
AP_R4 27C512
687N08 785X08
A_S6 6264SL
A_S4 6264SL
MWORKCS
MREAD1
MLWR
MREAD2
MUWR
MWRITE
A_L3 74LS138
address decoder
B_C6 74LS138
B_G4 74LS273
flip flops
VFLIP
HFLIP
PI[0..2]
VRAMCVF
EXRES
SYSFLAG
MCOMCS
COLORRAM
IOCS
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
DMACS
These components are on the video PCB.
A_M3.4B 74LS02
A_L4B 74LS74
A_M4B 74LS21
74LS07
A_J1A 74LS32
A_L4A 74LS74
A_M4C 74LS21
A_N3A 74LS139
FIXVRAMCS
VRAMCS
OBJCS
A_G6A 74LS00
A_K1A 74LS32
A_H4A 74LS08
A_H7A 74LS04
A_J1B 74LS32
A_K1D 74LS32
A_H7E 74LS04
A_H7B 74LS04
A_H7F 74LS04
A_L2B 74LS00
A_L2C 74LS00
A_M3A 74LS74
A_M3B 74LS74
A_H4D 74LS08
A_L1A 74LS07
A_L1D 74LS07
A_L1E 74LS07
68000
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
UDS
LDS
R/W
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
MA12
MA13
MA14
MA15
MA16
MA17
MA18
MA19
MA20
MA21
MA22
MA23
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
MD8
MD9
MD10
MD11
MD12
MD13
MD14
MD15
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Main CPU/
File: mcpu.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 2/17

SDB1 SSYSFLAG A_F6A 74LS74 SINT16EN EXRES

SDB0 SSYSFLAG A_H6B 74LS74 MINT EXRES

SDB2 SSYSFLAG A_G5A 74LS74 CHAPAGE

Only used on PWB(A)302190B
Goes to CN2:A23.
On PWB(A)302190A CN2:A23 is connected to GND.

A_M8 74LS138 address decoder
SA17 SA18 SA19 SAS SA22 → SCOMCS SWORKCS SSYSFLAG

A_T7 74LS138 CPU address decoder in bold
SA19 SA20 SA21 SA22 SAS SA23
$40 SOBJCS $48 SVRAMCS $50 $58 $60 $68 $70 $78 OCHARAMCS
A_M4A 74LS21 OCHAROMCS
74LS00 A_G6D
A_F6B 74LS74 VCHARAMCS
GND TIM3 called ZIP RAM in test screens

SA[15..23]

A_H6A 74LS74 GND SINT
A_H5A 74LS74 GND VBLANK
74LS00 A_G6C SCPUCLKD
A_H4B 74LS08

A_G6B 74LS00
SA23 SAS 74LS04 A_F7A
A_RA7 4.7k x 4

68000
CLK IPL0 IPL1 IPL2 BGACK BG BR FC0 FC1 FC2 VMA E VPA BERR DTACK HALT RESET AS UDS LDS R/W
A_J8 A1–A23 D0–D15

A[0..14]
A_L9 3-StateHexBuffers 74LS367
A_L10 74LS244
A_M10 buffers 74LS245 A->B
A_M9 buffers 74LS245 A->B
D[0..15]

A_N10 27C512 687N07 785P07
AP_R10 27C512 687L11
A_S10 27C512 687N12 785P13
A_U10 6264SL
A_N8 27C512 687N07 785P07
A_R8 27C512 687L10
A_S8 27C512 687N13 785P12
A_U8 6264SL

8R and 10R are not mounted when the GX687 board is used for gradius 2 with GX785 roms.

SAB[1..14] SDB[0..15]

In the original schematics
DMAINT, MODTACK and DMAINIT are the same

A_M3.4D 74LS02 SOBJCS DMAINIT A_V10B 74LS74 SODTACK
A_K1B 74LS32 SAS SA22
A_M3.4C 74LS02 SVRAMCS A_V10A 74LS74 VRAMDTACK
A_H8C 74LS07 CPURESD A_H8B 74LS07
A_S7A 74LS30 OCROMDTACK OCRAMDTACK
A_H7D 74LS04

A_J1C 74LS32 SUWR
A_J1D 74LS32 SLWR
A_H7C 74LS04 SREAD

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Slave CPU/
File: scpu.kicad_sch

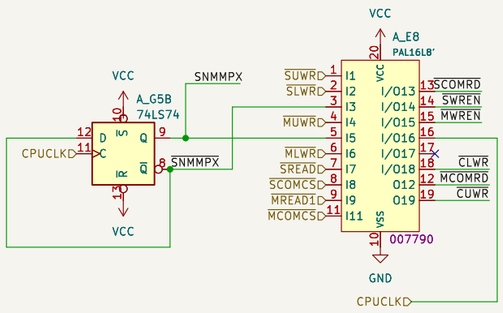
Title: Twin 16

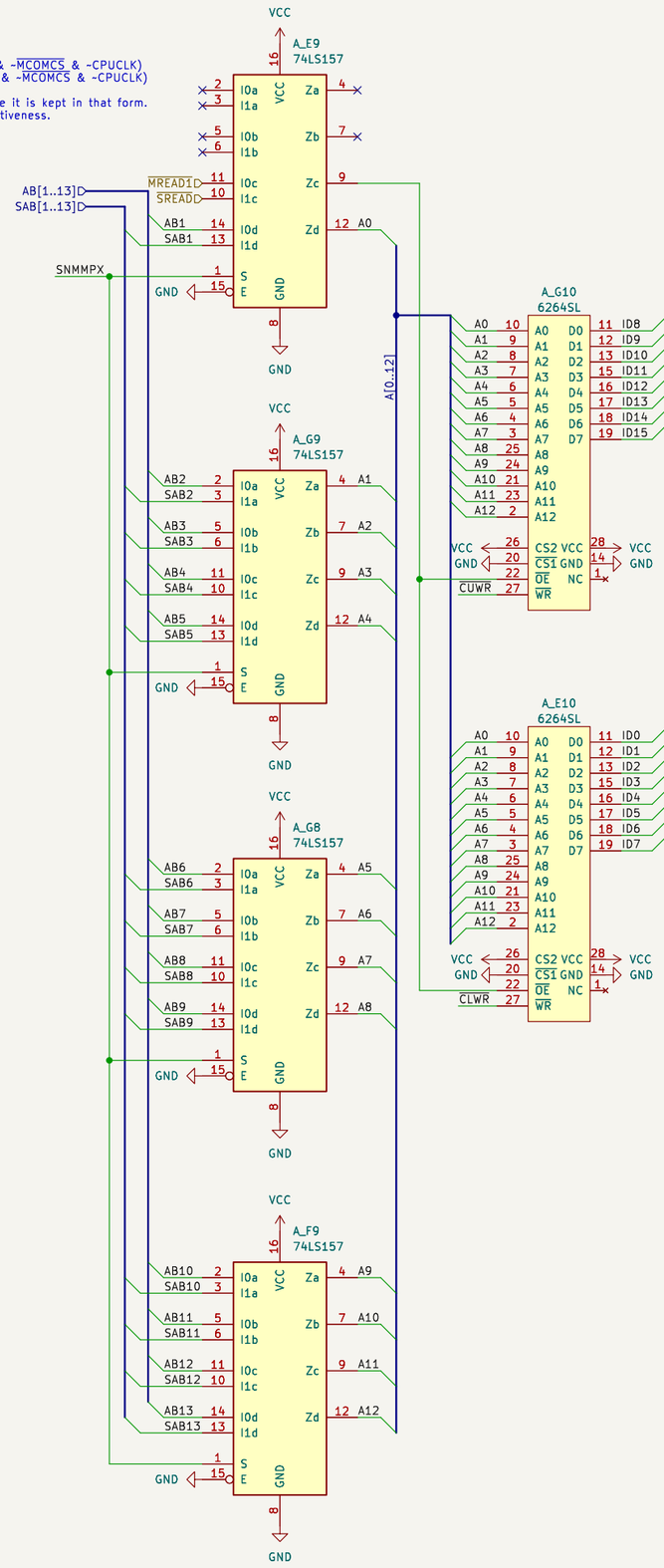
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 3/17

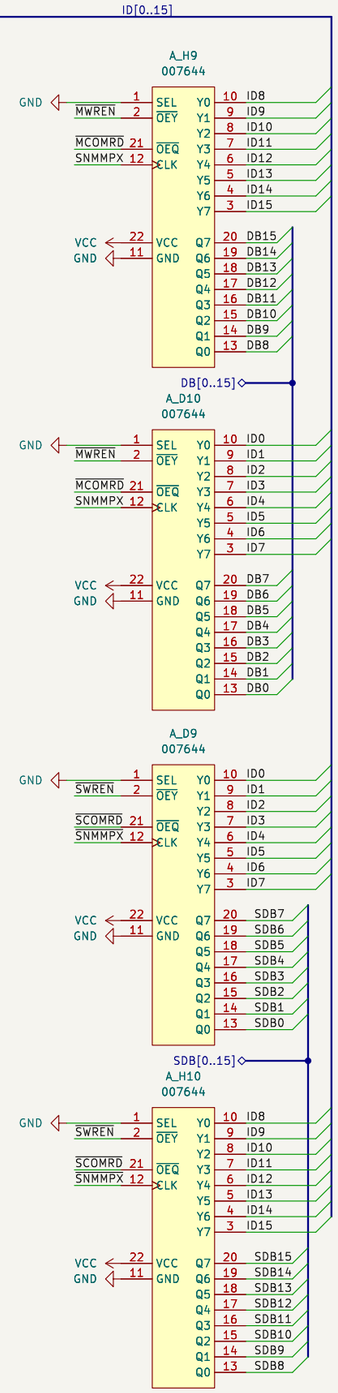
Konami 007790

$\overline{MCOMRD}$ = ~(~$\overline{MREAD1}$ & ~$\overline{MCOMCS}$) = $\overline{MREAD1}$ | $\overline{MCOMCS}$
$\overline{SCOMRD}$ = ~(~$\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{SWREN}$ = ~(~SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$)
$\overline{MWREN}$ = ~(~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$)
$\overline{CLWR}$ = ~(~$\overline{SLWR}$ & ~SNMMPX & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~SNMMPX & ~$\overline{MLWR}$ & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)
$\overline{CUWR}$ = ~(~$\overline{SUWR}$ & ~SNMMPX & SNMMPX & $\overline{SREAD}$ & ~$\overline{SCOMCS}$ & ~CPUCLK | ~$\overline{MUWR}$ & ~SNMMPX & $\overline{MREAD1}$ & ~$\overline{MCOMCS}$ & ~CPUCLK)

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form.
The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.







This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs
mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs.
If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

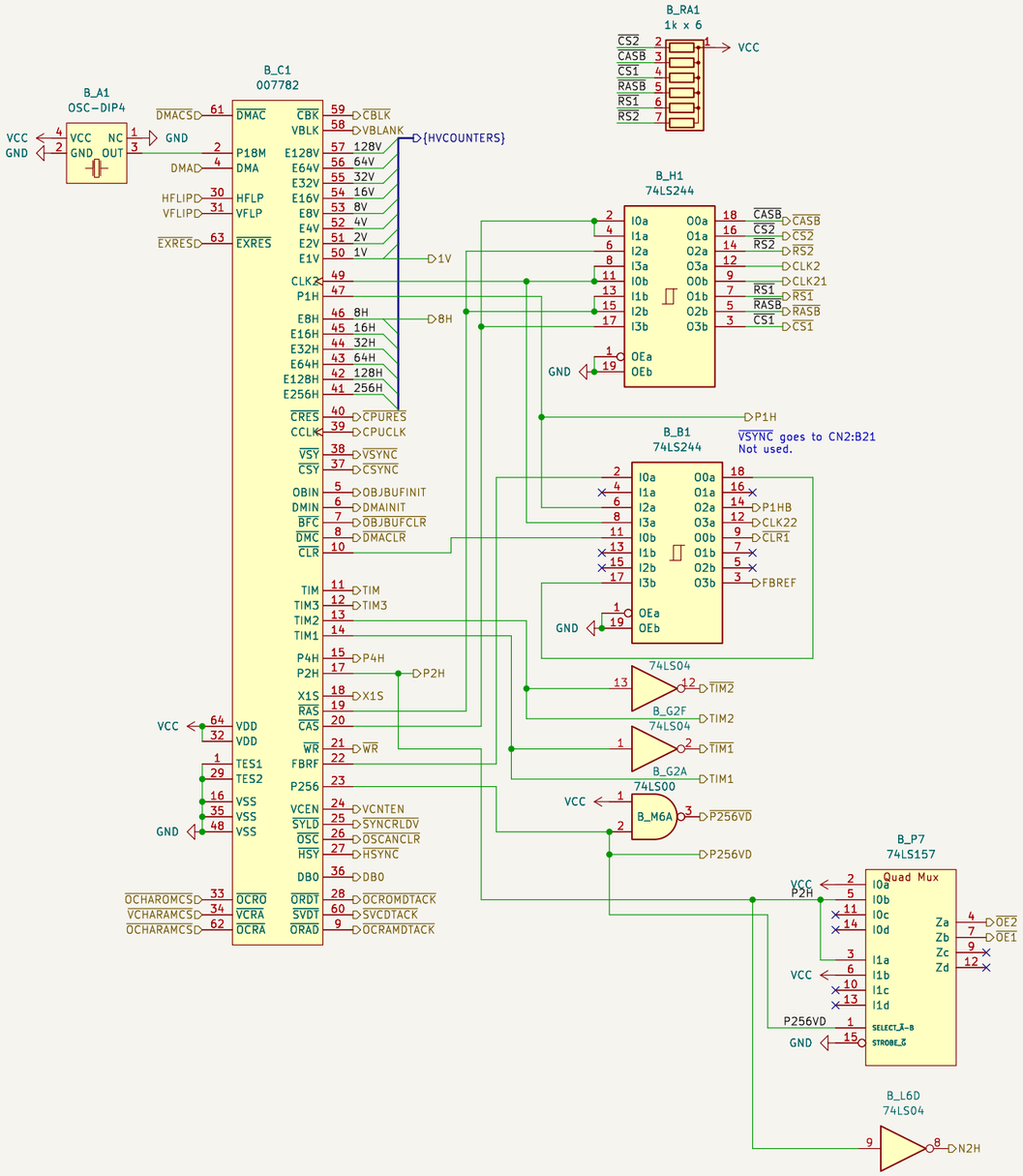
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Konami GX687 / GX785

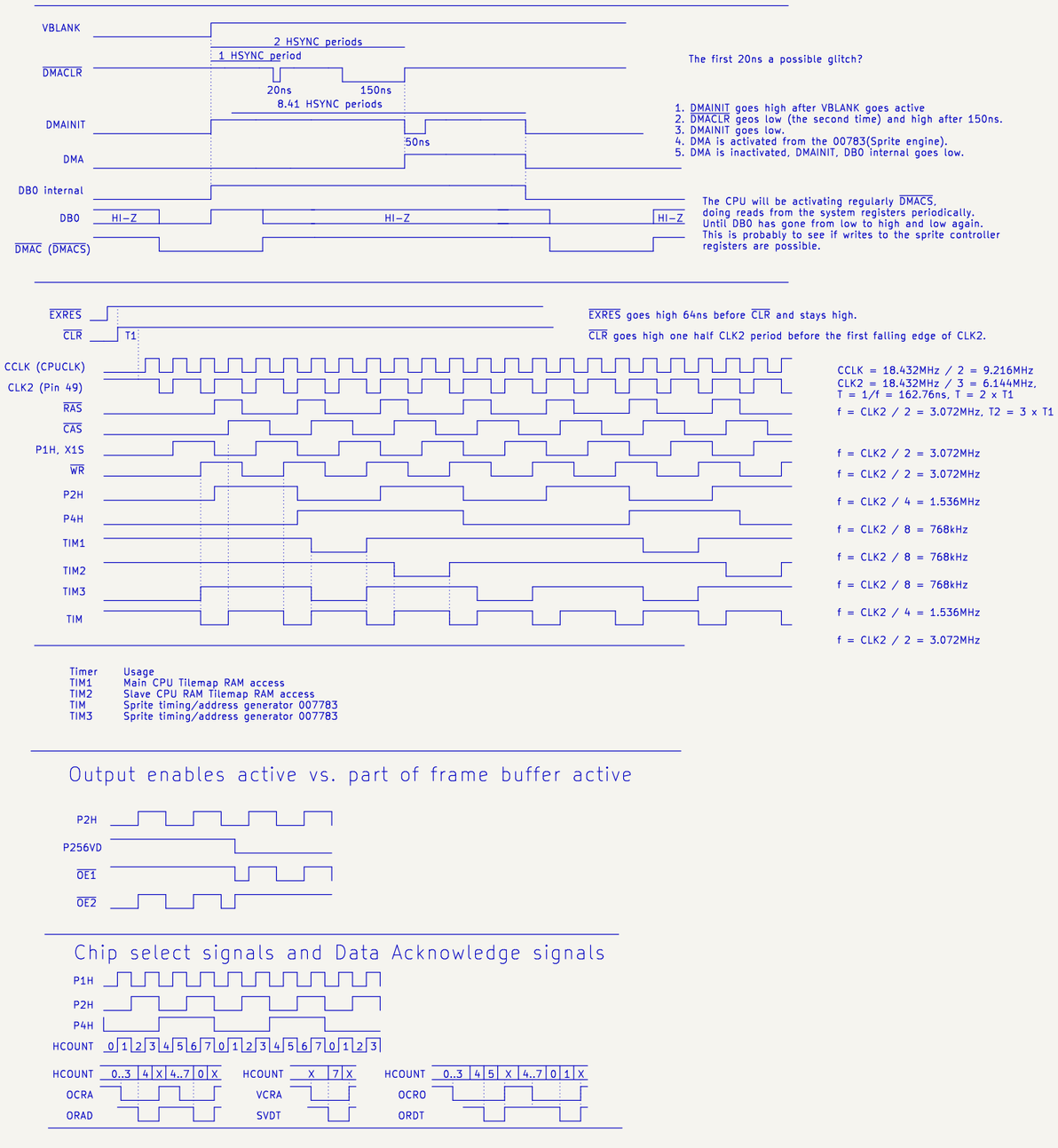
Sheet: /Shared Resources/
File: shared_resources.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 4/17



DMA Synchronization



Timer	Usage
TIM1	Main CPU Tilemap RAM access
TIM2	Slave CPU RAM Tilemap RAM access
TIM	Sprite timing/address generator 007783
TIM3	Sprite timing/address generator 007783

Output enables active vs. part of frame buffer active

Chip select signals and Data Acknowledge signals

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Konami GX687 / GX785

Sheet: /Timing/
File: timing.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 5/17

B_A2
74LS157
Quad Mux
AB[1..13]
AB12
AB13
GND
P2HD
GND
A11
A12
B_B2
74LS157
Quad Mux
AB8
AB9
AB10
AB11
16V
32V
64V
128V
P2HD
GND
A7
A8
A9
A10
B_B4B
74LS27
B_L6A
74LS04
FIXVRAMCSD
P2HD
MUWRD
B_C2
74LS157
Quad Mux
AB4
AB5
AB6
AB7
64H
128H
256H
8V
P2HD
GND
A3
A4
A5
A6
A[0..12]
B_B4A
74LS27
B_L6F
74LS04
FIXVRAMCSD
P2HD
MLWRD
B_D2
74LS157
Quad Mux
MREAD2D
AB1
AB2
AB3
8H
16H
32H
P2HD
GND
OE
A0
A1
A2
10ns delay
50ns delay from OE active
to output data valid
B_A3
6264SL
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
VCC
GND
CS2
CS1
OE
WR
VCC
GND
NC
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
D14
D15
D[0..15]
B_B3
6264SL
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
VCC
GND
CS2
CS1
OE
WR
VCC
GND
NC
D0
D1
D2
D3
D4
D5
D6
D7
7ns delay
B_J1A
74LS86
VCC
P4HD
P4H
20ns delay
B_C4
74LS174
D9
D10
D11
D12
D13
D14
VCC
D0
D1
D2
D3
D4
D5
Cp
Mr
hex flip flop
Q0
Q1
Q2
Q3
Q4
Q5
Palette, HFLIP and VFLIP is valid for
eight pixels at a time and becomes valid
on the rising edge of P4H
B_C3
74LS174
VCC
hex flip flop
7ns delay
74LS86
VCC
P1HBD
B_J1C
P1HB
P2HD
P4H
HFLIPD
B_J2A
B_J1B
B_J2C
B_J2D
B_H2C
B_H2B
B_H2D
B_D3
74LS174
1V
2V
4V
D0
D1
D2
hex flip flop
B_D4
74LS174
D3
D4
D5
D6
D7
D8
hex flip flop
B_F3
27C128
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
GND
CE
OE
PGM
VPP
VCC
D0
D1
D2
D3
D4
D5
D6
D7
687M14/L14/N14
785H14/G14
Two pixels color data is output
on each rising edge of P1HB.
The current pixel data is multiplexed
with P1HB (and HFLIP).
B_E2
74LS273
D0
D1
D2
D3
D4
D5
D6
D7
flip flops
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
FX20
FX21
FX22
FX23
FX10
FX11
FX12
FX13
FX[20..23]
FX[10..13]
B_F2
74LS157
Quad Mux
FX10
FX11
FX12
FX13
FX20
FX21
FX22
FX23
GND
Za
Zb
Zc
Zd
74LS86
P1HBD
HFLIP
B_J1D
B_H2A
The total delay for a pixel is seven pixel clocks
from when P2H and OE goes high and low respectively.
B_G3
74LS273
B_H3
74LS273
B_H4
74LS273
D0
D1
D2
D3
D4
D5
D6
D7
flip flops
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
CLK21D
VCC
FX0
FX1
FX2
FX3
FX4
FX5
FX6
FX7
FX[0..7]
FX[0..3]: Color
FX[4..7]: Palette
B_D5
74LS245
DB15
DB14
DB13
DB12
DB11
DB10
DB9
DB8
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
buffers
A->B
CE
MREAD2D
D15
D14
D13
D12
D11
D10
D9
D8
B_E5
74LS245
DB7
DB6
DB5
DB4
DB3
DB2
DB1
DB0
D7
D6
D5
D4
D3
D2
D1
D0
B_C5A
74LS32
FIXVRAMCSD
P2HD
{HVCOUNTERS}
DB[0..15]
RAM
A[0..5] Horizontal position 0..504 in steps of 8
A[6..10] Vertical position 0..248 in steps of 8
A[11.12] Tilemaps? Up to 4.
ROM
A[0..1] Horizontal position 0..6 in steps of 2
A[2..4] Vertical position 0..7
A[5..13] Tile number
D[0..8]: Tile number
D[9..12]: Palette
D13: HFLIP
D14: VFLIP
3. Data out latched
into DFF's.
4. DFF's data
out valid
6. DFF's data
out valid
CLK21
P1H
P1HB
P2H
P4H
1
2
3
4
5
6
7
8
2. Data out from
RAM valid
4. FX[0..7] Data out valid
5. Palette and color values
latched into DFF's 3C and 2E.
1. Addresses and OE
latched into RAM. This condition sets the start:
P2H & P4H = 1.
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Fixed Layer/
File: fixed_layer.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
Rev:
KiCad E.D.A. 8.0.9
Id: 6/17

ZIP RAM

256kB tile RAM
16-bit access by CPU
32-bit access by video

64k x 4bit

CPUCTL flag must be set in 007779
to get the VA bus driven by the CPUs

VD[0..12]: Tile number
VD[13..15]: Palette Color

Data latched from the RAM:
Edge Tilemap
T1–T2 1
T2–T3 2

It is output 2 pixel clocks later to the ROMs.

When VA is switched to output the bus
is updated for each P1H cycle.

The vertical counters are not the same value
when they are offset with the scroll registers.

Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 –> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 –> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

Outputs of RAM active when P4H is low, i.e during T1 and T2.

SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

This is from jammarcade.net

Konami 007644

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs.
If SEL is LOW then the outputs will only change state on a rising edge CLK.

The 007781 is a shift register. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that. It reads 32 bits / 4 bits / pixel = 8 pixels at a time.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Tile Layers/
File: tile_layers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 7/17

B_K1
007783

B_L3
6264SL

B_M3
6264SL

B_K3
007784

74LS245 B_J5
74LS245 B_G5
74LS245 B_H5
74LS245 B_F5

B_K5
007785

74LS157 Quad Mux B_K7
74LS157 Quad Mux B_L7
74LS157 Quad Mux B_N6
74LS157 Quad Mux B_M7
74LS157 Quad Mux B_N7

B_C5B 74LS32
B_J8A 74LS32
B_K6A 74LS32
B_K6C 74LS32

B_H11A 74LS74

B_P1 PAL16L8'
007793

B_RA2 2.2k x 8
B_RA3 2.2k x 8

AB[1..13]
SAB[1..14]
OA[0..12]
CRW[0..3]
OD[0..15]
DB[0..15]
SDB[0..15]
SA[15..20]
O[0..19]
VD[7..0]
HD[8..2]
HJ[0..1]
CX[0..3]
C[0..3]
SPA[0..17]
VPD8
{RW1}
{RW2}

On PWB(A) 302190A CHAPAGE (CN2:A23) is connected to GND

Each unparsed sprite (object) occupies 40 words in object RAM. Only the first 8 words of each object data is used. The memory range for unparsed data is up 17FFh. That leaves room for 153 (153 x 40 −1 = 17E7h) objects per frame.

007784 Sprite parser

− subtracts the scroll value for the sprites from current coordinates for the sprites.
− When DMA goes high. The 007784 will read 8 words, if bit OD15 of the first word is high then BEFLAG goes high and stays high until the sprite RAM has processed the first four words and possibly updated the four following words at the high end of the RAM. It will compress the eigth words into four words and also subtract the x− and y−scroll values from the x and y coordinates. It will write the four words at the high end of the sprite RAM.
− Addresses at the low end goes from 0x0000..0x0007 and the following cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when the signals OBJSCRV and OBJSCRH go low.

007793

CLK21′ = CLK21

LRW = ~(DMAONS & OBJBUFINIT & CLK21′ | BEFLAG & DMA & P1H & DMAONS & CLK21′ & (~8H & HFLIP | 8H & ~HFLIP) ~LWR & ~DMAINIT)

URW = ~(DMAONS & OBJBUFINIT & CLK21′ | BEFLAG & DMA & P1H & DMAONS & CLK21′ & (8H & ~HFLIP | ~8H & HFLIP) | ~UWR & ~DMAINIT)

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG & DMA
3. ~DMAINIT

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite Table/
File: sprite_table.kicad_sch

Title: Twin 16

Size: A3	Date: 2024−09−08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 8/17

Graphics Tile Data (ROM and RAM)

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.

Konami 007788

The inverted ~P2H is written as N2H.

$\overline{OCRDL}$ = ~(~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWL}$ = ~(~P2H & $\overline{SREAD}$ & SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{OCRDH}$ = ~(~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$ & $\overline{OCHAROMCS}$ | ~$\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHAROMCS}$)
$\overline{OCWH}$ = ~(~P2H & $\overline{SREAD}$ & ~SAB1 & ~$\overline{OCHARAMCS}$)
$\overline{VCWENL}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & SAB1)
$\overline{VCRDL}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & SAB1)
$\overline{VCRDH}$ = ~(~$\overline{VCHARAMCS}$ & ~$\overline{SREAD}$ & ~SAB1)
$\overline{VCWENH}$ = ~(P2H & P4H & ~$\overline{VCHARAMCS}$ & $\overline{SREAD}$ & ~SAB1)
$\overline{P2HP4H}$ = ~(P2H & P4H)
$\overline{N2HP4H}$ = ~(~P2H & P4H)

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

128kB RAM
for dynamic tiles

Vertical flipped ROM's?

Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Sprite ROM and RAM/
File: sprite_rom_and_ram.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 9/17

Frame buffer 1

Frame buffer 2

2 x 256kB frame buffers
Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

Not certain which of the parts 6KB,6UB and 6UD are used where. Some errors in the original schematics.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | -WR | -VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

OCD[0:3]: Pixel color
OCD[4:7]: Palette

Frame buffer 1 Palette color

Frame buffer 2 Palette color

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Konami GX687 / GX785

Sheet: /Sprite Frame Buffers/
File: sprite_frame_buffers.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	Rev:
KiCad E.D.A. 8.0.9		Id: 10/17

Components are on Video PCB

Components are on Main PCB

object layer:
0000: Transparent
1111: Shadow
else: Opaque

B_A11D 74LS32 · B_A11B 74LS32 · B_A11C 74LS32

FX2 12 · FX3 13 · FX1 9 · FX0 10 → 11, 8 → 6 FX

B_J8D 74LS32 · B_J8C 74LS32

1C3 12 · 1C2 13 · 1C1 9 · 1C0 10 → 11, 8 → 6 1C

B_J11 GAL22V10 007789

PI[0..2] → PI0 1 · PI1 2 · 2C0 3 · PI2 4 · OCD0 5 · OCD1 6 · OCD2 7 · OCD3 8 · 1C6 9 · 2C6 10 · I1/CLK 11 · I13

I/O/Q: 23 SHADOW · 22 · 21 SELB · 20 SELA · 19 2C3 · 18 2C2 · 17 2C1 · 16 · 15 · 14

VCC 24 · GND 12

{SEL}

B_H9 74LS153 Mux 4–1: GND 6 · VCC 4 · GND 3 · GND 1 · VCC 10 · 11 · 12 · GND 13 · GND 15 · SELA 14 · SELB 2 · 7 COLOR8 · 9 COLOR9

B_H8 74LS153 Mux 4–1: 2C6 6 · 1C6 5 · OCD6 4 · FX6 3 · GND 1 · VCC 10 · GND 11 · 12 · 13 · GND 15 · SELA 14 · SELB 2 · 7 COLOR6 · 9 COLOR7

B_H10 74LS153 Mux 4–1: 2C4 6 · 1C4 5 · OCD4 4 · FX4 3 · GND 1 · 2C5 10 · 1C5 11 · OCD5 12 · FX5 13 · GND 15 · SELA 14 · SELB 2 · 7 COLOR4 · 9 COLOR5

B_J9 74LS153 Mux 4–1: 2C2 6 · 1C2 5 · OCD2 4 · FX2 3 · GND 1 · 2C3 10 · 1C3 11 · OCD3 12 · FX3 13 · GND 15 · SELA 14 · SELB 2 · 7 COLOR2 · 9 COLOR3

B_J10 74LS153 Mux 4–1: 2C0 6 · 1C0 5 · OCD0 4 · FX0 3 · GND 1 · 2C1 10 · 1C1 11 · OCD1 12 · FX1 13 · GND 15 · SELA 14 · SELB 2 · 7 COLOR0 · 9 COLOR1

2C[0..6] · 1C[0..6] · OCD[0..7] · FX[0..7]

COLOR[0..9]

```
Konami 007789

i11 (1C6) and i5 (PI2) is not used

~o15 = OCD0 & OCD1 & OCD2 & OCD3 | ~OCD0 & ~OCD1 & ~OCD2 & ~OCD3

o16 = OCD0 & OCD1 & OCD2 & OCD3

SELA = FX +
       /PI1 &   PI0 & 1C +
       /PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & 1C +
       /PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & 1C +
        PI1 &  OCD0 &  OCD1 &  OCD2 &  OCD3 & /2C0 & /2C1 & /2C2 & /2C3 +
        PI1 & /OCD0 & /OCD1 & /OCD2 & /OCD3 & /2C0 & /2C1 & /2C2 & /2C3

SELB =/FX & OCD0 & OCD1 & OCD2 & OCD3 +
       /FX & /OCD0 & /OCD1 & /OCD2 & /OCD3 +
       /FX & PI0 & /PI1 & 1C +
       /FX & PI0 &  PI1 & 2C6 & 2C0 +
       /FX & PI0 &  PI1 & 2C6 & 2C1 +
       /FX & PI0 &  PI1 & 2C6 & 2C2 +
       /FX & PI0 &  PI1 & 2C6 & 2C3
o21.oe = vcc

/SHADOW = /FX & /PI1 & OCD0 & OCD1 & OCD2 & OCD3 +
          /FX & /PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /1C  +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C6 +
          /FX &  PI0 & OCD0 & OCD1 & OCD2 & OCD3 & /2C0 & /2C1 & /2C2 & /2C3
```

AB[1..12]

A_U6 74LS273 flip flops: COLOR2 3 · COLOR3 4 · COLOR4 7 · COLOR5 8 · COLOR6 13 · COLOR7 14 · COLOR8 17 · COLOR9 18 · CLK2D 11 · VCC 1 · D0 D1 D2 D3 D4 D5 D6 D7 · Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 · Cp · Mr · 2 CB2 · 5 CB3 · 6 CB4 · 9 CB5 · 12 CB6

A_T6 74LS157 Quad Mux: AB9 2 · AB10 5 · AB11 11 · AB12 14 · 3 · 6 · 10 · 13 · GND · COLORRAM 1 · GND 15 · I0a I0b I0c I0d · I1a I1b I1c I1d · SELECT_A–B · STROBE_G · Za 4 C9 · Zb 7 C8 · Zc 9 C7 · Zd 12 C6

A_U7 74LS174 hex flip flop: CBLK 3 · 4 · 6 · 11 · COLOR0 13 · COLOR1 14 · CLK2D 9 · VCC 1 · D0 D1 D2 D3 D4 D5 · Cp · Mr · Q0 Q1 Q2 Q3 Q4 Q5 · 2 · 5 · 7 · 10 · 12 CB0 · 15 CB1

CBLK

CB[0..6]

A_U1 007327: C6 20 · C7 19 · C8 18 · C9 17 · AB8 29 · AB7 30 · AB6 31 · AB5 32 · AB4 33 · AB3 34 · AB2 35 · AB1 37 · RA11 RA10 RA9 RA8 A7 A6 A5 A4 A3 A2 A1 A0 · CB6 22 · CB5 23 · CB4 24 · CB3 25 · CB2 26 · CB1 27 · CB0 28 · DB7 3 · DB6 4 · DB5 5 · DB4 6 · DB3 7 · DB2 8 · DB1 9 · DB0 10 · D7 D6 D5 D4 D3 D2 D1 D0 · VCC 12 · GND 11 · GND 16 · CLK2 40 · MREAD1 38 · COLORRAM 21 · 1 · 2 · 39 · CLK R/W SEL CCS CWR BLK

CB[6..9]

AB1 3 · A_F7B · 4 · 74LS04

R 15 → RED · G 14 → GREEN · B 13 → BLUE

A_R3 82 · A_R4 82 · A_R5 82

DB[0..7]

VCC · GND

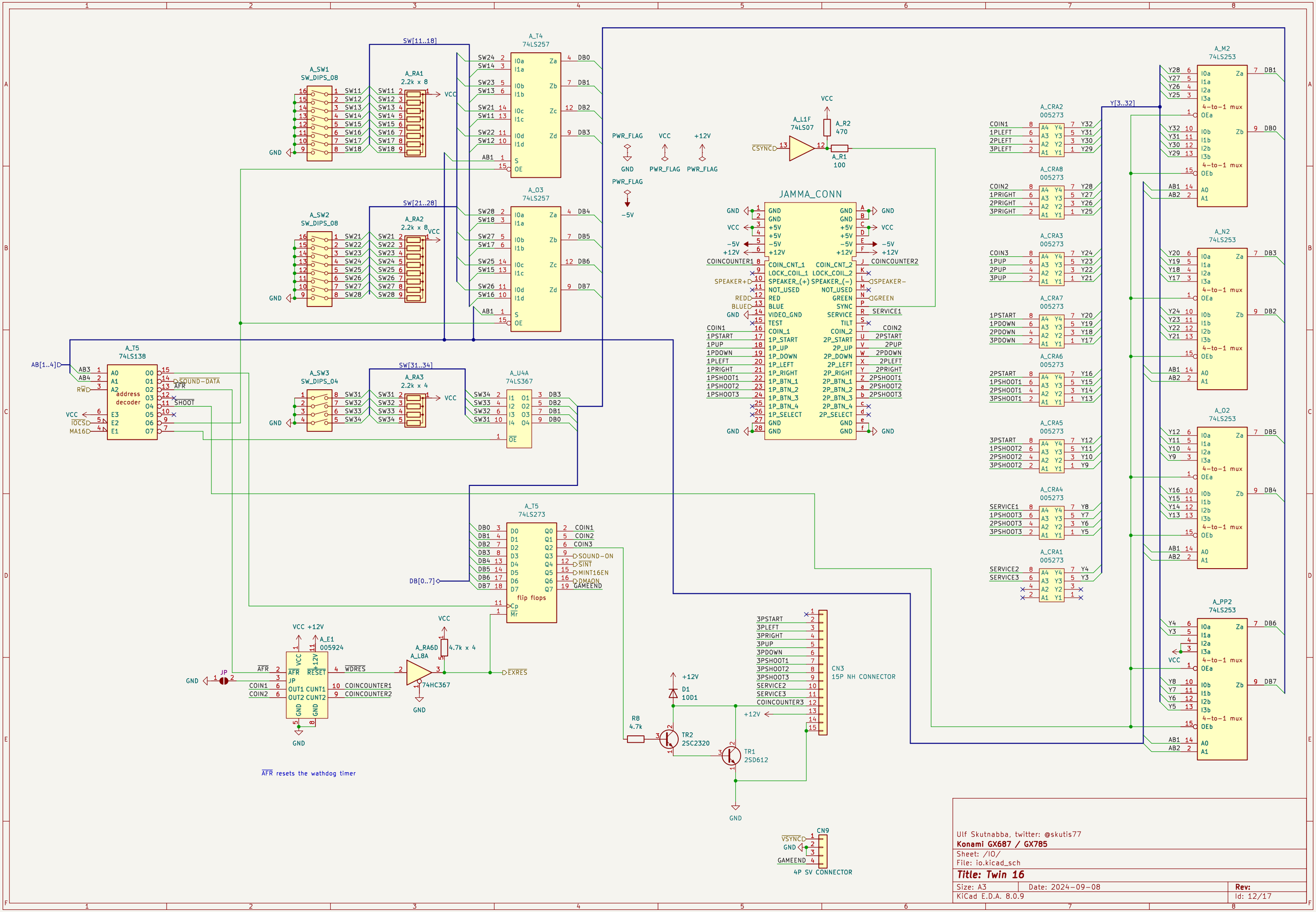
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Color mixer/
File: color_mixer.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 11/17



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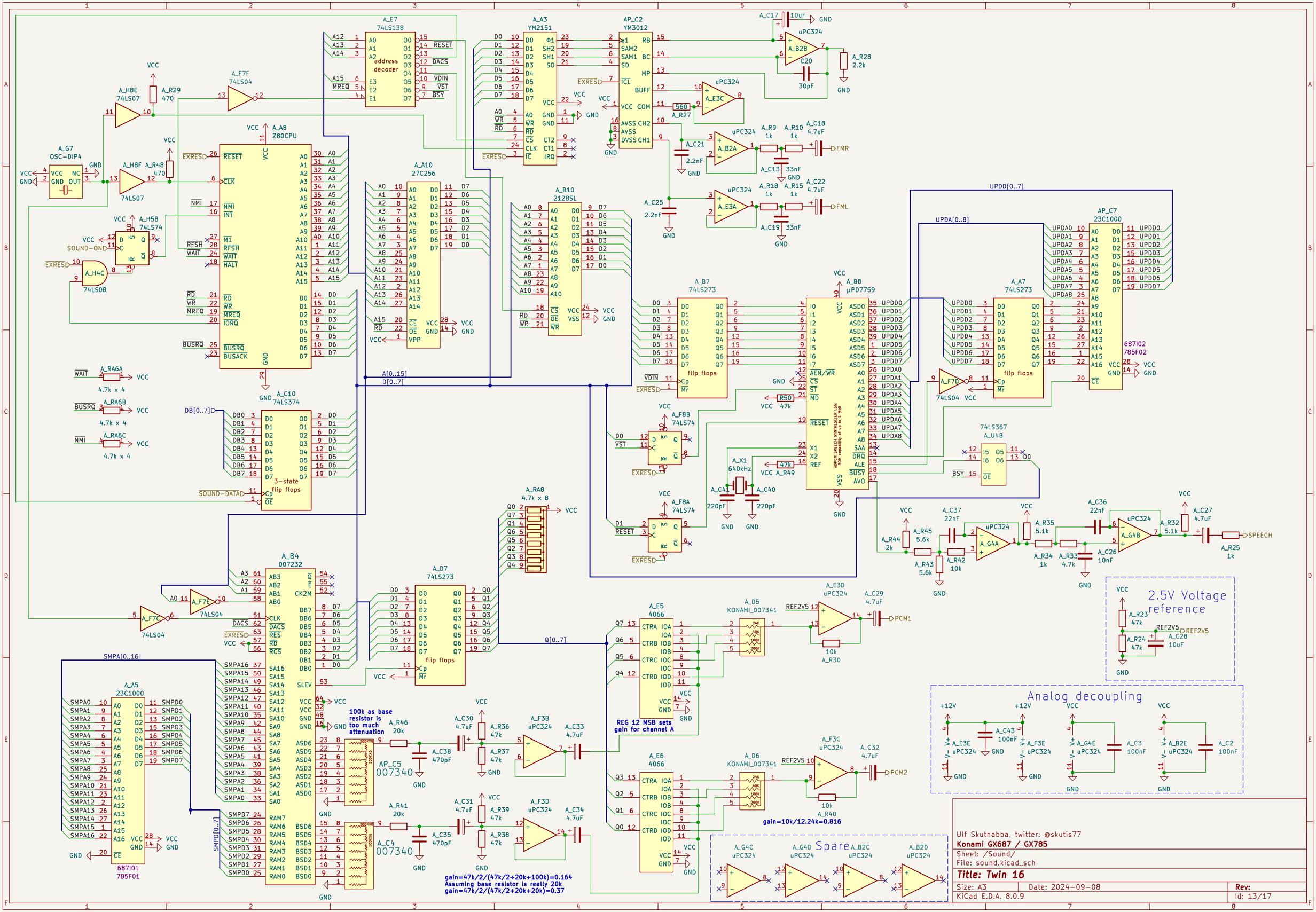
Konami GX687 / GX785

Sheet: /IO/

File: io.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 12/17

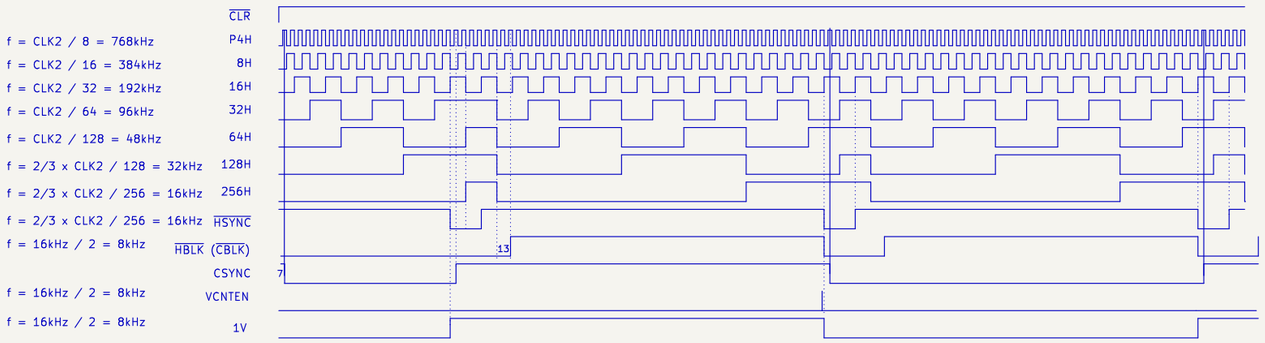

A_E7
74LS138
address decoder
A_A3
YM2151
AP_C2
YM3012
A_F7F
74LS04
A_H8E
74LS07
A_R29
470
A_G7
OSC-DIP4
A_H8F
74LS07
A_R48
470
A_A8
Z80CPU
A_A10
27C256
A_B10
2128SL
A_H5B
74LS74
A_H4C
74LS08
A_B7
74LS273
A_B8
µPD7759
A_A7
74LS273
AP_C7
23C1000
A_C10
74LS374
3-state flip flops
A_F8B
74LS74
A_F8A
74LS74
A_X1
640kHz
A_RA6A
4.7k x 4
A_RA6B
4.7k x 4
A_RA6C
4.7k x 4
A_RA8
4.7k x 8
A_B4
007232
A_D7
74LS273
A_E5
4066
A_E6
4066
A_D5
KONAMI_007341
A_D6
KONAMI_007341
AP_C5
007340
A_C4
007340
A_A5
23C1000
687I01
785F01
687I02
785F02
A_F7E
74LS04
A_F7C
74LS04
A_F7D
74LS04
74LS367
A_U4B
A_C17
10uF
A_B2B
uPC324
A_C20
30pF
A_R28
2.2k
A_E3C
uPC324
A_R27
560
A_C21
2.2nF
A_B2A
uPC324
A_R9
1k
A_R10
1k
A_C18
4.7uF
A_C13
33nF
A_C25
2.2nF
A_E3A
uPC324
A_R18
1k
A_R15
1k
A_C22
4.7uF
A_C19
33nF
FMR
FML
A_R46
20k
A_C38
470pF
A_R36
47k
A_R37
47k
A_F3B
uPC324
A_C33
4.7uF
A_R41
20k
A_C35
470pF
A_R39
47k
A_R38
47k
A_F3D
uPC324
A_C34
4.7uF
A_F3C
uPC324
A_C32
4.7uF
A_R50
47k
A_R49
47k
A_C41
220pF
A_C40
220pF
A_R44
2k
A_R45
5.6k
A_R43
5.6k
A_C37
22nF
A_R42
10k
A_G4A
uPC324
A_R35
5.1k
A_R34
1k
A_R33
4.7k
A_C36
22nF
A_C26
10nF
A_G4B
uPC324
A_R32
5.1k
A_C27
4.7uF
A_R25
1k
SPEECH
A_E3D
uPC324
A_C29
4.7uF
A_R30
10k
PCM1
A_R40
10k
PCM2
2.5V Voltage reference
A_R23
47k
A_R24
47k
A_C28
10uF
REF2V5
Analog decoupling
A_E3E
uPC324
A_C43
100nF
A_F3E
uPC324
A_G4E
uPC324
A_C3
100nF
A_B2E
uPC324
A_C2
100nF
Spare
A_G4C
uPC324
A_G4D
uPC324
A_B2C
uPC324
A_B2D
uPC324
A[0..15]
D[0..7]
Q[0..7]
SMPA[0..16]
SMPD[0..7]
UPDA[0..8]
UPDD[0..7]
DB[0..7]
REG 12 MSB sets gain for channel A
100k as base resistor is too much attenuation
gain=10k/12.24k=0.816
gain=47k/2/(47k/2+20k+100k)=0.164
Assuming base resistor is really 20k
gain=47k/2/(47k/2+20k+20k)=0.37
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Sound/
File: sound.kicad_sch
Title: Twin 16
Size: A3
Date: 2024-09-08
KiCad E.D.A. 8.0.9
Rev:
Id: 13/17

To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

FMRD
SPEECHD
PCM1D
PCM2D
FMLD
SPEECHD
PCM1D
PCM2D
REF2V5D
REF2V5D
A_R17 1k
A_R12 7.5k
A_R11 5.1k
A_R19 5.1k
A_R16 1k
A_R22 7.5k
A_R31 5.1k
A_R20 5.1k
A_E3B uPC324
A_F3A uPC324
A_R13 12k
A_R21 12k
+12V
+12V
A_R52 3.3k
A_R53 4.7uF
A_C15 4.7uF
A_C16
A_R14 10k
A_R26 10k
CN6 4P NH CONN
CN7 4P NH CONN
GND
GND
APOT_R53 1k
APOT_R52 1k
A_C11 100nF
A_C12 100nF
GND
GND
GND
GND
+12V
A_C9 100uF
A_D1 MB3722
ACAP_C3 1000uF
GND
GND
GND
ACAP_C10 100uF
A_C14 10uF
GND
IN-A
IN+A
IN+A
IN-B
VCC
GNDIN
GNDOUT
BYPASS
BSA
OUTA
BSB
OUTB
A_C8 100uF
A_C7 100uF
ACAP_C2 1000uF
A_C6 100nF
A_R6 1
A_C5 100nF
A_R7 1
A_C1 1000uF
GND
GND
GND
SPEAKER-
SPEAKER+
CN8 4P NH CONN

Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch
Title: Twin 16
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 14/17

Horizontal signals

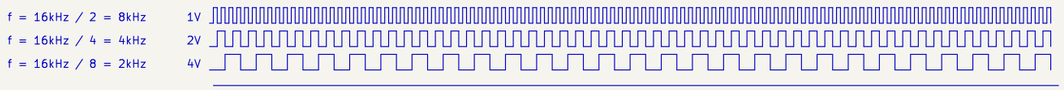


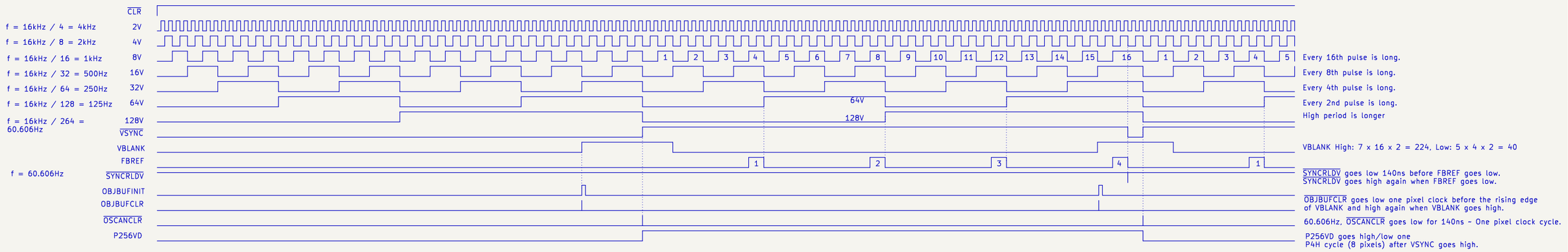
The first HSYNC is at HCOUNT 175 after reset.

– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

Vertical signals





Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer

VBLANK High: 7 x 16 x 2 = 224, Low: 5 x 4 x 2 = 40

SYNCRLDV goes low 140ns before FBREF goes low.
SYNCRLDV goes high again when FBREF goes low.

OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.

60.606Hz, OSCANCLR goes low for 140ns ~ One pixel clock cycle.

P256VD goes high/low one
P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

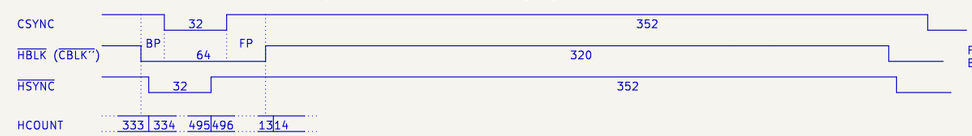
Horizontal and vertical synch timing diagrams

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

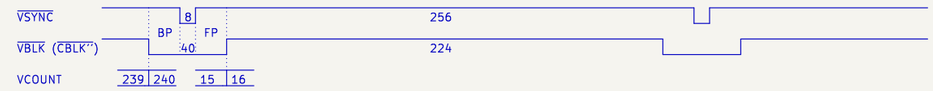
Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



Front Porch (FP) 24 px,
Back Porch (BP) 8 px

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

CBLK'' is at the output of color mixer 007327.
CBLK' is shifted inside the 007327 one pixel clock.

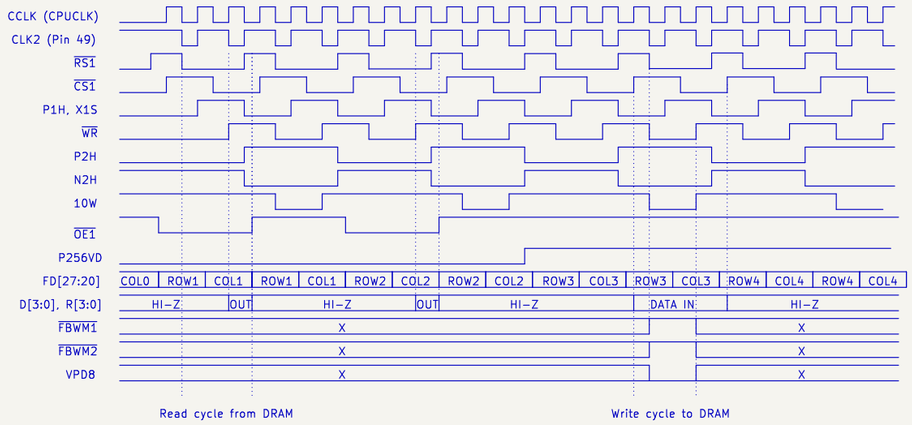
Ulf Skutnabba, twitter: @skutis77

Konami GX687 / GX785

Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
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The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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Konami GX687 / GX785

Sheet: /Sprite timing diagrams/
File: sprite_timing_diagrams.kicad_sch

Title: Twin 16

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.9		Id: 16/17

D_E1
74LS174
DB'8 3
DB'9 4
DB'10 6
DB'11 11
DB'12 13
D0
D1
D2
D3
D4
D5
Q0
Q1
Q2
Q3
Q4
Q5
14
15
2 DB''8
5 DB''9
7 DB''10
10 DB''11
12 DB''12
9
Cp
CPURES' 1
hex flip flop
Mr
DB''[12..8]
D_F1A
74LS04
MA16' 1
2
D_G1B
74LS32
IOCS' 4
5
6 RAM
D_F1B
74LS04
AB'10 3
4
D_G1A
74LS32
1
2
3
D_G1C
74LS32
9
AB'10 10
8 RWEN
D_F1C
74LS04
DB''12 5
6
D_F2D
74LS32
12
13
11 HRAMEN
74LS32
2
1
3 LRAMEN
D_E2A
D_G2
6264SL
AB'1 10
AB'2 9
AB'3 8
AB'4 7
AB'5 6
AB'6 5
AB'7 4
AB'8 3
AB'9 25
DB''8 24
DB''9 21
DB''10 23
DB''11 2
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
D0
D1
D2
D3
D4
D5
D6
D7
11 DB'0
12 DB'1
13 DB'2
15 DB'3
16 DB'4
17 DB'5
18 DB'6
19 DB'7
VCC
26
CS2 VCC
CS1 GND
OE NC
WR
28 VCC
14 GND
1
RAMEN 20
22
27
MLWR' 9
HRAMEN 10
D_F2C
8
74LS32
HRAMEN 1
MREAD2' 2
D_F2A
3
74LS32
D_I2
6264SL
AB'1 10
AB'2 9
AB'3 8
AB'4 7
AB'5 6
AB'6 5
AB'7 4
AB'8 3
AB'9 25
DB''8 24
DB''9 21
DB''10 23
DB''11 2
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
D0
D1
D2
D3
D4
D5
D6
D7
11 DB'8
12 DB'9
13 DB'10
15 DB'11
16 DB'12
17 DB'13
18 DB'14
19 DB'15
VCC
26
CS2 VCC
CS1 GND
OE NC
WR
28 VCC
14 GND
1
RAMEN 20
22
27
MUWR' 4
HRAMEN 5
D_F2B
6
74LS32
AB'[15..0]
74LS245
DB0 2
DB1 3
DB2 4
DB3 5
DB4 6
DB5 7
DB6 8
DB7 9
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
18 DB'0
17 DB'1
16 DB'2
15 DB'3
14 DB'4
13 DB'5
12 DB'6
11 DB'7
buffers
MREAD2' 1
RAM 19
A->B
CE
D_C1
74LS245
DB8 2
DB9 3
DB10 4
DB11 5
DB12 6
DB13 7
DB14 8
DB15 9
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
18 DB'8
17 DB'9
16 DB'10
15 DB'11
14 DB'12
13 DB'13
12 DB'14
11 DB'15
buffers
MREAD2' 1
RAM 19
A->B
CE
D_D1
74LS245
AB1 2
AB2 3
AB3 4
AB4 5
AB5 6
AB6 7
AB7 8
AB8 9
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
18 AB'1
17 AB'2
16 AB'3
15 AB'4
14 AB'5
13 AB'6
12 AB'7
11 AB'8
buffers
VCC 1
GND 19
A->B
CE
D_C2
74LS245
AB9 2
AB10 3
MREAD2D 4
MLWRD 5
MUWRD 6
MA16D 7
IOCSD 8
CPURESD 9
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
18 AB'9
17 AB'10
16 MREAD2'
15 MLWR'
14 MUWR'
13 MA16'
12 IOCS'
11 CPURES'
buffers
VCC 1
GND 19
A->B
CE
D_D2
AB'[10..1]
AB[10..1]
DB[15..0]
CN1
Conn_02x20_Odd_Even
GND 1
3
2 GND
4
DB0 5
DB2 7
DB4 9
DB6 11
DB8 13
DB10 15
DB12 17
DB14 19
6 DB1
8 DB3
10 DB5
12 DB7
14 DB9
16 DB11
18 DB13
20 DB15
AB1 21
AB3 23
AB5 25
AB7 27
AB9 29
22 AB2
24 AB4
26 AB6
28 AB8
30 AB10
MREAD2D 31
MUWRD 33
IOCSD 35
CPUCLKD 37
VCC 39
32
34 MLWR
MA16 36
CPURES
38 VBLANK
40 VCC
(Not used)
(Not used)
CN5 is on main board.
CN1 is on sub board PWB451803 and
mates to CN5.
D_F1D
74LS04
9
8
D_F1E
74LS04
11
10
D_F1F
74LS04
13
12
D_G1D
74LS32
12
13
11
VCC
14
VCC
GND
7
GND
D_F1G
74LS04
VCC
VMEM
D_U1
1
VCC
OUT
GND
2
M51944BSL
3 RAMEN
D_R1
10k
If VCC > 4.25V
OUT is activated
GND
D_D11
VCC 1S1588 VMEM
D_D12
1S1588
D_BT1
3V
GND
VMEM
PWR_FLAG
VMEM is connected to VCC
on the RAM IC's.
GC903
Ulf Skutnabba, twitter: @skutis77
Konami GX687 / GX785
Sheet: /Battery Powered RAM/
File: bpram.kicad_sch
Title: Twin 16
Size: A3
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